

Android Architecture:-

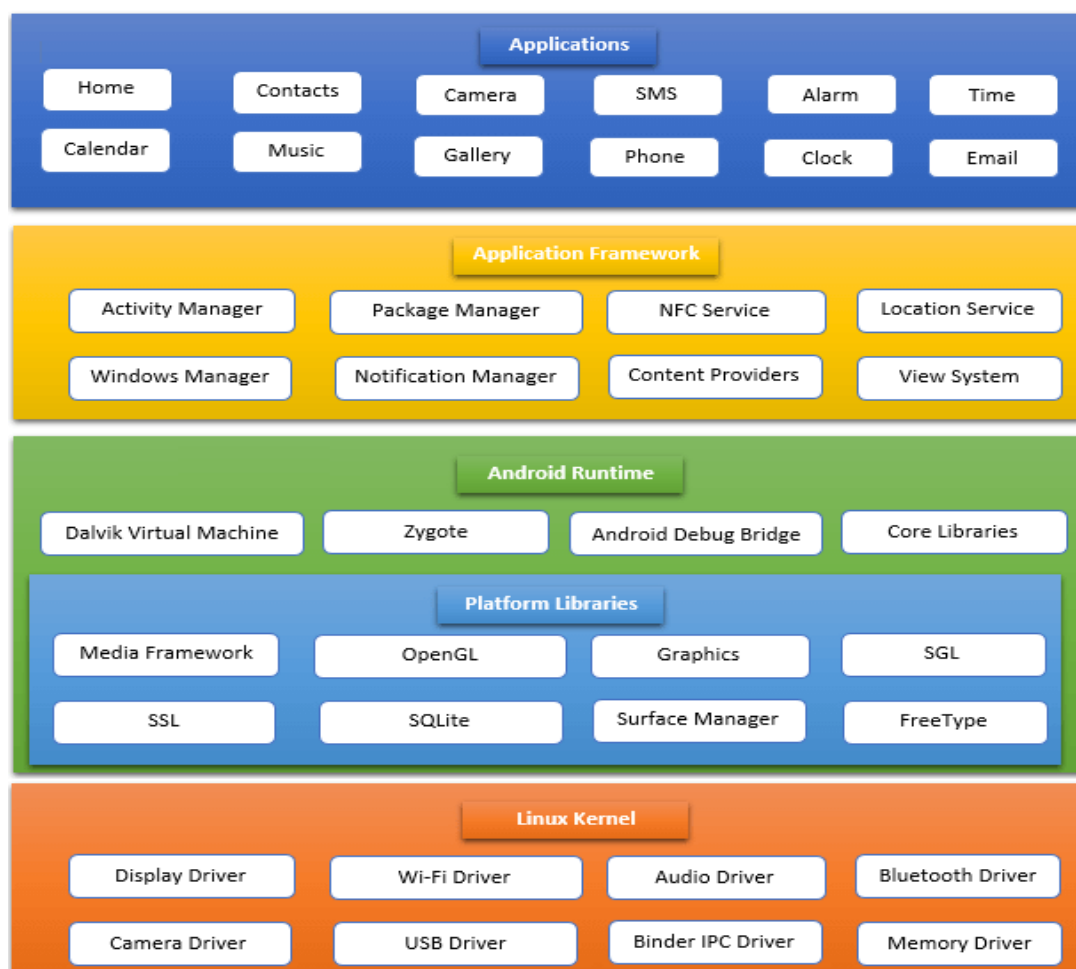
1. What is Android Architecture?

Android Architecture is the layered structure of the Android operating system that explains how an Android application communicates with the operating system, system services, and device hardware.

Android Architecture tells us how Android works internally and how an application interacts with the mobile device.

Android is organized into different layers. Each layer performs a specific job and communicates with the layer above or below it.

2. Android Architecture Diagram



3. Layer 1 — Applications

This is the top layer.

It contains applications that users interact with.

Examples

Phone

YouTube

WhatsApp

Instagram

Your Android Application

When you develop an Android application in Android Studio, your application eventually runs at this layer.

Example

Suppose you create:

Food Delivery App

The user sees:

Food Delivery

Pizza

Burger

French Fries

ORDER NOW

The user interacts with the application layer.

4. Layer 2 — Android Application Framework

This is the layer that provides important services and APIs to application developers.

Instead of developers directly controlling hardware, Android provides ready-made framework components.

example, your application wants to:

- Open another screen
- Display a notification
- Access location
- Manage application packages
- Display UI

Android Framework provides services for these operations.

Important Application Framework Components

1. Activity Manager

Manages Activities and their lifecycle.

For example:

LoginActivity



HomeActivity



ProfileActivity

It helps Android manage these screens.

2. Window Manager

Manages application windows and screens.

It deals with things such as:

Application windows

Dialogs

Screen display

Window positioning

3. Notification Manager

Used to display notifications.

Example:



New Message

You received a new message.

Applications use Android's notification system to show such alerts.

5. Layer 3 — Android Runtime

Android Runtime is the environment used to execute Android application code.

The modern Android runtime is called:

ART — Android Runtime

Older versions of Android used:

Dalvik Virtual Machine (DVM).

ART replaced Dalvik in Android 5.0 and later.

6. Native C/C++ Libraries

Android also contains many important native libraries, primarily implemented in C/C++.

These libraries provide low-level functionality used by Android and applications.

Important examples include:

SQLite

Used for local database functionality.

Example:

Student Records

OpenGL ES

Used for graphics.

Examples:

2D Graphics

Animations

Media Framework

Handles multimedia operations.

Examples:

Image

Recording

Playback

SSL/TLS Libraries

Support secure network communication.

For example:

HTTPS

Secure API Communication

7. Layer 4 — Hardware Abstraction Layer (HAL)

HAL stands for:

Hardware Abstraction Layer

This layer provides a standard interface between Android's higher-level software and device-specific hardware implementations.

Why is this needed?

Different companies manufacture different devices.

example:

Samsung

OnePlus

Their hardware can be different.

But Android applications should not need to understand the internal implementation of every camera, microphone, sensor, etc.

HAL helps provide a common interface.

Example: Camera

Suppose your application wants to use the camera.

The simplified communication is:

Your Camera App



Android Framework



Camera HAL



Camera Driver



Camera Hardware

8. Layer 5 — Linux Kernel

The Linux Kernel is the lowest layer of Android Architecture.

It communicates with the device hardware and provides core operating-system functionality.

Main responsibilities

1. Device Drivers

Drivers allow Android to communicate with hardware.

Examples:

Camera Driver

Display Driver

2. Memory Management

The kernel manages RAM.

For example:

App A → Memory

App B → Memory

App C → Memory

It manages how memory is allocated and reclaimed.

3. Process Management

The kernel manages processes running on the device.

Example:

WhatsApp Process

Android System Process

Your App Process

Android Activity Lifecycle

1. What is an Activity?

An Activity is a component of an Android application that generally represents a screen with which the user can interact.

Examples:

LoginActivity → Login Screen

MainActivity → Home Screen

ProfileActivity → Profile Screen

SettingsActivity → Settings Screen

An Activity does not remain active forever. Android moves it through different states depending on what the user is doing.

This sequence of states is called the Activity Lifecycle.

2. What is Activity Lifecycle?

Activity Lifecycle is the sequence of states and callback methods through which an Activity passes from creation to destruction.

Android provides lifecycle callback methods that allow developers to perform specific tasks when an Activity:

- Is created
- Becomes visible
- Comes to the foreground
- Loses focus
- Becomes invisible
- Comes back
- Is destroyed

3. Activity Lifecycle Methods

The main lifecycle methods are:

onCreate()

onStart()

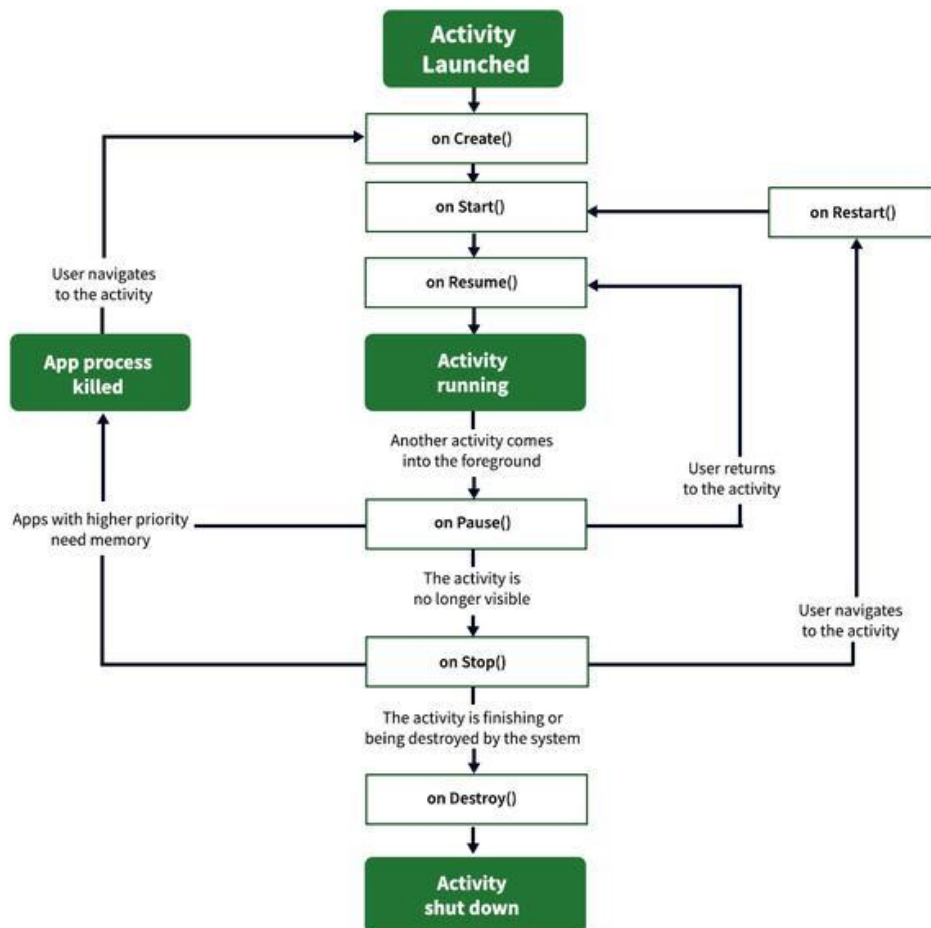
onResume()

onPause()

onStop()

onRestart()

onDestroy()



Activity Lifecycle in Android

**Lifecycle
Method****One-line Description****onCreate()**

Called when the Activity is first created; used to initialize the UI and data.

onStart()

Called when the Activity becomes visible to the user.

onResume()

Called when the Activity comes to the foreground and the user can interact with it.

onPause()

Called when the Activity loses focus or is partially covered by another screen.

onStop()

Called when the Activity is no longer visible to the user.

onRestart()

Called when a stopped Activity is about to become visible again.

onDestroy()

Called when the Activity is being destroyed or finished.